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IS HYDROGEN SULFIDE SAFETY PROCEDURE

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1.0 PURPOSE

This Procedure outlines the requirements for safe operations and conduct of personnel in the probable presence of a Hydrogen Sulfide (H₂S) environment and provides guidance for the minimum standards of breathing air systems and associated equipment on all worksites within QatarEnergy-IS operations.

2.0 SCOPE AND APPLICABILITY

This Procedure applies to all QatarEnergy-IS facilities, equipment and personnel involved with work activities that are exposed to or require protection from potential exposure to H₂S within QatarEnergy operations.

3.0 ROLES AND RESPONSIBILITIES

HSE MANAGER

- Assist in the review and approval of Self Contained Breathing Apparatus (SCBA) purchased or rented/leased by QatarEnergy-IS.
- Ensure a training program for H₂S protection is developed and made available to the affected operation.
- Ensure an initial audit is conducted of contractor's equipment for compliance to this Procedure prior to mobilization.

MANAGERS/ LEAD

- Ensure this Procedure is communicated within their group.
- Ensure training has been provided for the use and maintenance of SCBAs, as required.
- Ensure H₂S Contingency Plan is developed and topics identified within are addressed.
- Ensure a risk assessment is conducted addressing the potential for exposure to H₂S on the worksite.

OFFSHORE SUPERVISORS

- Head of Maintenance is responsible to ensure that the necessary planned maintenance routines / programs are being efficaciously implemented which should be appropriately set out in this procedure as well.
- Occupation Health and Paramedics – as part of coordination of air quality tests and of compilation of records / reports of air quality tests for the attention of relevant parties onshore.
- Industrial Hygienist – verifications of air quality parameters through the relevant reports submitted by paramedics and / or breathing apparatus technicians and occupational exposure limits for hydrogen sulphide gas concentrations
- Facility Services Supervisors – collection of hydrogen sulphide gas training cards for lifeboat capsules and distribution of adequately bump tested personal hydrogen sulphide gas detectors as necessary.
- Offshore Installation Superintendents – overall accountability in terms of the efficacy of implementation of the standing procedural stipulations.



4.0 PROCEDURAL STEPS

4.1 GENERAL

When H₂S is contained in the reservoir, anything that has been downhole has the potential to bring H₂S to the surface either in oil, drilling fluids, produced water, gas, etc.

An un-ignited H₂S release is the predominant risk to personnel. The emergency response philosophy is predicated on the fact that any process or drilling release may have H₂S associated with it.

H₂S is known to exist at QatarEnergy-IS offshore locations and as such all personnel shall be trained to the H₂S worker level in the use of a breathing apparatus, and the specific Procedures and controls which are in place at that worksite.

4.2 H₂S CONTINGENCY PLAN

Each QatarEnergy-IS offshore unit shall have in place a site specific H₂S Contingency Plan / Emergency Response Plan covering toxic gas release incidents.

The purpose of the Hydrogen Sulfide Contingency is to provide an organized plan of action for alerting and protecting the public following an accidental release of a potentially hazardous volume of hydrogen sulphide. The plan shall be a logical step-by-step approach to an emergency and coordinated with the Workplace Emergency Response and Crisis Communication plans. The requirements for and content of the Contingency Plan may be covered by regulations (Refer to Appendix D for process flow diagram).

The plan shall include, at a minimum:

- a) Instructions for alerting employees and the public in case of an emergency.
- b) Procedure for requesting assistance and follow-up action to remove the public from the area of exposure.
- c) A call list of internal and external people to notify in the event of an emergency.
- d) A list of contacts for contractors and service companies available for emergency response activities including type of equipment.
- e) Plot of area showing radius of exposure, location of Public Areas, location of evacuation routes, location of safety equipment and telephones.
- f) List of names and telephone numbers of residents within the area of exposure and the person responsible for any Public Area.
- g) Provision for advance briefing of public within an area of exposure. This briefing shall include information on the Hazards and characteristics of hydrogen sulfide, possible sources, instructions for reporting a leak and the manner in which the public will be notified of an emergency.
- h) Detailed operating Procedures to be followed in an emergency including specific job assignments for personnel.
- i) Detailed remedial Procedures to be followed in an emergency
- j) Criteria for determining when protective equipment such as clothing, SARs and SCBAs, gas detectors, etc. is required to be used by personnel.
- k) Consideration of weather, differences in terrain, vegetation and seasonal changes that might cause an increase in Hazardous conditions or abnormal dispersion pattern.

4.3 SELF CONTAINED BREATHING APPARATUS (SCBA)

Appropriate training shall be given for site specific respiratory protective equipment and shall at a minimum include:



- a) Location of self-contained breathing apparatus.
- b) Location of spare cylinders.
- c) Site specific issues; environmental hazards, skill levels, detection levels, monitoring devices, etc.
- d) Situations/conditions requiring SCBA use.
- e) Limitations and capabilities of positive pressure/full face respirators.
- f) Brand/Model/Size of respirators available.
- g) Annual fit testing and medical evaluation are mandatory for all personnel who are members of the installations emergency response teams. Refer to Respiratory Protection Procedure (IS-HSE-PRC-013) for further guidance.

4.4 ESCAPE BREATHING APPARATUS (EBA)

Personnel shall go from PS1Q to the jacket they are working on PS1 without the need to carry an EBA unit. Upon reaching their intended jacket they will take an EBA from the designated EBA rack that is located on each of the jackets. If an EBA unit is not available on said jacket rack then the Max POB for this jacket has been reached and the individual shall go back to PS1Q to obtain an EBA unit to take to his work site.

For Jackets PS1C, PS1D and PS1K the EBA units must be within immediate reach of personnel at all times. For all other locations EBAs must be within work location.

The OIS, PS1 Safety Officer and Operators are not required to take EBA units from the main deck racks, instead their EBA units are located on all levels of jackets in yellow SCBA boxes. However while they are on PS1C, PS1D and PS1K they are still required to have EBA units within Immediate reach.

Note- EBA units shall be available at muster stations on PS1Q and Marinha for personnel in accommodation.

The above process is summarized in Figure 1: EBA use on PS1.

FIGURE 1: EBA USE ON PS1



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4.5 H₂S TRAINING

- a) H₂S training is mandatory for all personnel entering an QatarEnergy-IS offshore.
- b) H₂S training shall be conducted by an approved training provider. H₂S training shall focus on the skills and knowledge required to work safely in a H₂S environment and shall meet the requirements of OPITO
- c) H₂S Trainers shall be fluent in English.

4.4.1 Proof of Certification of H₂S Training

Each person shall have available, an H₂S training certification card from an OPITO or QatarEnergy approved trainer, inclusive of:

- Name of individual.
- Signature of training Approver
- Date of training (including expiration date).
- Name and address of training provider.
- H₂S training certification is only valid for a period of one-year (2-years for QatarEnergy).

4.4.2 H₂S Training Records

Documentation detailing the content of the H₂S Training Program shall be maintained and available, including; Name of Person, Training Provider, Validity along with copy of the card. The records shall also be available in Dawinci, as applicable.

4.6 DRILLS

Muster drills shall be conducted at the facility during normal duty hours and shall be performed by all personnel. The drills shall consist of a dry-run performance of personnel activities related to assigned jobs.

Each drill shall be critiqued and any new considerations documented. Records of each drill and a critique summary shall be kept for review and documentation purposes.

For Offshore Units - A minimum of one drill shall be carried out every seven (7) days (alternating fire/abandon ship and H₂S evacuation drills are acceptable) to verify in particular that all POB are familiar with donning and breathing through the BA sets.

For Halul PWHF Operations, which works under the QatarEnergy Halul Terminal Emergency Response Procedure, a frequency of one drill per month is acceptable.

4.7 FACIAL HAIR

All personnel who are required to wear BA masks shall be clean shaven. (Refer to APPENDIX A).

4.8 WIND INDICATORS

Wind indicators or windsocks shall be strategically located throughout the offshore unit and shall be positioned to allow viewing by a majority of personnel on the facility.



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4.9 ANCILLARY EQUIPMENT

The following ancillary equipment shall be available for conducting the Breathing air quality checks:

- a) Breathing Air Quality Test Kit - Flow meter and sample tubes (to test to CSG Grade D Breathing Air -APPENDIX B) - for units with cascade systems.
- b) Gas Sample Pump with certified tubes:-
 - i. CO detector tubes, range 5-50 ppm
 - ii. CO₂ detector tubes, range 100-11,500 ppm
 - iii. H₂S detector tubes, range 10-3000 ppm
 - iv. H₂S detector tubes, range 0.1-4.0 %
 - v. H₂S detector tubes, range 1-300 ppm
 - vi. H₂O detector tubes, range 0.1-30 mg/m³
 - vii. SO₂ detector tubes, range 1-200 ppm
 - viii. SO₂ detector tubes, 0.05 - 8 %
- c) High pressure fill lines for SCBAs
- d) Helicopter landing prohibition flag - for units with helidecks

4.10 SELF CONTAINED BREATHING APPARATUS

- a) All SCBAs/EEBAs shall be fitted with positive pressure mask mounted demand regulators and shall conform to a recognized oil and gas industry standard, such as US National Institute of Occupational Safety and Health (NIOSH) or equivalent.
- b) All SCBAs/EEBAs on location shall have same connections as per Cascade system
- c) All SCBAs/EEBAs on location shall be compatible with cascade system and maintenance capabilities.
- d) All breathing apparatuses shall be capable of being connected to a breathing air cascade system by means of a stainless steel quick connect fitting.
- e) SCBA/EEBAs Stainless steel quick connect fittings shall be compatible with both QatarEnergy and Contractor equipment, the type and manufacturer shall be determined during the Risk Assessment process.
- f) Dependent on the type of 30 min SCBA or 20min EEBAs (PS1) 15min EEBAs (Remotes) supplied, a 12" low pressure manifold pigtail hose may be required to be attached to the SCBA. This is in addition to the low-pressure manifold hose at the muster station manifolds.
- g) All 15 min/20min EEBAs shall be fitted with a quick refill connection.
- h) Each SCBA/EEBA shall be subjected to a monthly inspection and be subjected to a recertification, inspection, maintenance Procedure as per the manufacturer recommendation or as conditions warrant.
- i) The SCBA/EEBA inspection program for all SCBA equipment, hoses, regulators, bottles, etc. shall be documented, evidence of such programs and inspections shall be kept and be available for review.
- j) SCBAs and EEBAs consolidated total shall be provided to supply at least one hundred and twenty five percent (125%) of the personnel on board. The risk assessment and H₂S Contingency plan shall determine the actual number and distribution of 15 minute/20 minute and 30-minute SCBAs.

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- k) SCBAs supplied to key emergency response personnel shall be equipped with voice amplifiers.
- l) Lifejackets shall be of a compatible design to allow them to be quickly and easily donned while wearing an SCBA.

4.11 GAS DETECTION EQUIPMENT

- a) Each Manned QatarEnergy-IS offshore unit shall have a fixed multi-channel H₂S and Combustible Gas detection system capable of digital readout at a central control panel located within a 24 hour manned area; it shall be equipped with audio and visual alarm indicators, together with portable and digital readout for H₂S and Combustible detectors.
- b) Fixed H₂S and Combustible Gas Detection Systems shall be supplied and installed by a professional engineer or engineering company approved by QatarEnergy-IS. Guidance shall be from the CGA or an international equivalent and reviewed by QatarEnergy-IS.
- c) Fixed H₂S sensors shall be set to activate at a reading of 10 ppm for H₂S on PS1 and 25 ppm for H₂S on Remotes -
- d) Fixed SO₂ sensors, where fitted, shall be set to activate at a maximum reading of 2 ppm - Low Level Alarm (Visual Alarm &, Local audible alarm), 5 ppm - High Level Alarm (Visual Alarm & Audible GPA).
- e) NB. Gas sensor settings can be set to activate below these levels dependent on the installations risk assessment requirements but shall never exceed the maximum readings stated.
- f) Fixed H₂S Sensors shall be calibrated at least quarterly
- g) Personal H₂S detectors (intrinsically safe), capable of sensing a minimum H₂S concentration of 5 ppm, are required for personnel at the work site for use when personnel are required to access production jackets/process areas where no fixed detection /alarm system exists. These devices are to be electronic, and capable of emitting an audible alarm.

4.12 SITE SPECIFIC REQUIREMENTS

4.12.1 Vessels & Barges

Fixed Gas Detection Equipment:

Continuous monitoring type H₂S & Combustible gas detectors shall be located at each of the following locations as a minimum:

- Upper & lower accommodation air intakes - port & starboard of vessel, (4) plus
- 1 sensor on stern of vessel total number of 5 minimum.

Visual & Audible Alarms:

Automatic H₂S & Combustible Gas explosion proof alarms (audible & visual) shall be located at the following locations as a minimum:

Vessel's Bridge, Main Deck

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Personal H₂S Gas Detectors:

When vessel/barge is within 500m zone - Personal H₂S detectors are worn by all crew involved in day-to-day operational and maintenance activities).

4.12.2 Halul Island PWHF

Fixed Gas Detection Equipment:

Continuous monitoring type H₂S & Combustible gas detectors are located in the pump pad area, with hydrocarbon detection located in the HVAC inlet of the office block.

Personal H₂S detectors are worn by all crew working in Operational

Visual & Audible Alarms:

Automatic H₂S & Combustible Gas explosion proof alarms (audible & visual) shall be strategically placed throughout all process areas of the station.

4.12.3 PS1

Fixed Gas Detection Equipment:

Continuous monitoring type H₂S & Combustible gas detectors shall be strategically placed throughout all process areas of the station.

Personal H₂S Gas Detection:

Personal H₂S detectors are worn by all crew working in Operational areas

Visual & Audible Alarms:

Automatic H₂S & Combustible Gas explosion proof alarms (audible & visual) shall be strategically placed throughout all process areas of the station, and on each level of the accommodation.

Breathing Air Cascade System:

- a) Shall be equipped with a comprehensive breathing air system that as a minimum meets the requirements of the OSHA standard ref. 29 CFR 1910.134 and the CGA bulletins No. G 7.0 and 7.1. Such breathing air systems shall provide grade D breathing air at the user outlets, as defined in the standard. These standards also specify the maintenance and care of breathing air compressors. These minimum standards shall be met at all times.
- b) The system shall comprise of Air compressors, regulators; skid mounted cascade storage bottles and strategically placed breathing air manifolds, complete with hose and stainless steel quick connect fittings that are compatible with both Company and Contractor supplied SCBAs.
- c) The breathing air system cascade bottles shall have the capability to supply a minimum of sixty (60) litres/min (2.1 cu ft/min) to each connection on the manifold breathing air system. The cascades shall be designed and sized appropriately to provide breathing air to the manifold stations for maximum number of personnel on board (POB) for not less than sixty (60) minutes.
e.g. Cascade Volume Requirements - e.g. - 90 POB x 60 mins x 2.1 cu ft = 11,340 cu ft Minimum Total Cascade Volume.

**IS HYDROGEN SULFIDE SAFETY PROCEDURE****DOC NO: IS-HSE-PRC-023****REV. 03****High Pressure Side:**

- a) Due to extreme temperature variations, consideration shall be given to thermal expansion of the gases in the system and system cascade bottles. Suitable protection shall be provided to ensure safe operation of the breathing air system.
- b) The system shall be capable of re-charging SCBAs/EEBAs to three thousand (3,000) PSI. The cascade bottle racks shall be equipped with flexible hose connections and quick fill connections capable of charging either Company or Vendor SCBA bottles after each use.
- c) Each cascade rack shall be connected to a high-pressure header, with Manual controls, allowing the transfer of compressed air between cascades. Due to this additional safety measure, no check valves shall be installed in the system other than the check valves installed as part of the various pressure regulators.
- d) Each cascade rack shall be fitted with a low-pressure alarm.
- e) The cascade racks shall be capable of being isolated manually in case of system supply line failure and capable of being individually operated to provide emergency breathing air to the respective areas.
- f) Distribution of the high pressure compressed Grade D breathing air to the high-pressure breathing air storage cascade system shall be with high-pressure stainless-steel tubing or high pressure rated anti-static hose and fittings of equivalent rating. The high-pressure tubing/hose shall be secured in place by means of stainless-steel tie wraps or equivalent at regular intervals.
- g) As a minimum, the high-pressure piping or hose shall be rated to the maximum pressure of the compressor plus ten per cent (10%).
- h) The high-pressure breathing air system and compressor shall be adequately protected from over pressure by pressure safety valves (PSVs) designed to operate at the system operating pressure plus ten per cent (10%).
- i) The Offshore units emergency control room shall be fitted with a pressure gauge that is tied into the high-pressure side of the breathing air system. This gauge will indicate overall system pressure available and will be used to determine emergency response action levels.
- j) Compressors shall be designed to provide Grade D breathing quality air at a specified rated pressure suitable to satisfy each cascade bottle's charging requirements. The main compressor shall be driven by an electric motor. A backup compressor, as capable as the main, shall be connected to the Unit's backup generator or shall be a free-standing compression engine unit. The system shall be capable of recharging all the SCBAs on site from zero psi to working pressure within two (2) hours.

Low Pressure Side:

- a) Breathing air manifold outlets shall be provided in areas where operations personnel may be required to use SCBA units, in order to carry out work tasks.
- b) Breathing air manifolds with sufficient number of breathing air outlets shall be supplied to accommodate:- $\text{max POB} \times 125\%$.
- c) Located as follows:
- d) $100\% \times \text{max POB}$ at Primary H₂S muster stations/evacuation points.
- e) $25\% \text{ max POB}$ distributed at Emergency I Control Work Areas
- f) A pressure gauge shall be installed at each manifold location to indicate the system pressure.



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- g) All cascade manifold fittings shall be stainless steel quick connect fittings, that are compatible with both Company and Contractor equipment, the type and manufacturer shall be determined during the Risk Assessment process.
- h) The low-pressure system shall be monitored by a visual and audible alarm system that alarms when the low-pressure system pressure is below 25% of its normal operating pressure.
- i) The low-pressure hoses, fittings and manifolds shall be protected from over pressurization by strategically placed PSVs set at no more than one hundred ten percent (110%) of normal system operating pressure, e.g. 110 psig on a 100 psig system.
- j) Where permanent breathing air cascade systems are installed, a site-specific schematic of the breathing air system layout is to be included. (APPENDIX C)

Commissioning of Breathing Air Cascade:

- a) All breathing air system pressure relief valves shall be certified before installation in the breathing air system.
- b) All crews shall demonstrate competency at a fire/abandon ship drill and an H₂S evacuation drill to the satisfaction of an QatarEnergy-IS representative. It is the Contractor's responsibility to ensure that these drills are witnessed by an QatarEnergy-IS representative.

H₂S Technician/ Maintenance Personnel

- a) Shall be trained and experienced in the maintenance, minor repair and instructing of others in the use of all breathing air system equipment provided. H₂S technicians shall be certified by the equipment manufacturer to the highest maintenance level certification achievable for field maintenance.
- b) Shall also be trained in the daily, weekly and annual care and maintenance of the breathing air compressors and shall be able to carry out testing of the stored air to ensure that it meets CGA grade D breathing air specifications.
- c) Shall operate the breathing air system compressors and shall ensure that the system cascades are fully charged at all times, all cascade bottles are in the fully open position, pipe work is free of leaks and compressors are set to operate in the Manual position.
- d) Shall conduct Breathing air quality testing, monthly at a minimum, which satisfy the parameters set by the Compressed Gas Association in CGA G7.1 'Commodity Specification for Breathing Air' APPENDIX B.
- e) Breathing air quality tests shall be conducted by sampling air at both the compressor and at the manifolds.
- f) Breathing air shall only be made when it has been confirmed by tri-meter that the air supply at the compressor inlet has been tested.
- g) All maintenance work carried out on the breathing air cascade system shall be controlled under the installations Integrated Safe System of Work (ISSOW) system.

4.12.4 MODUs & MOUs

Fixed Gas Detection Equipment:

Continuous monitoring type H₂S & Combustible gas detectors shall be located at each of the following locations as a minimum:



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- a) **MODUs** - Bell nipple, Drill Floor, Shakers, Mud Pits, Living Quarters (Port Side), Living Quarter (Starboard Side), above or around flow lines, **Monkeyboard**, Well Head Jacket. The system shall have the capacity to extend to the farthest point to which the worksite's personnel shall be required to work on any QatarEnergy-IS production jacket. **SO₂ detectors shall be placed on Monkeyboard & Drill Floor**
- b) **Well Service MOUs** - 2 strategically placed on main deck, air intakes to accommodation, Well Head Jacket. Helideck. The system shall have the capacity to extend to the farthest point to which the worksite's personnel will be required to work on any QatarEnergy-IS production jacket.

Personal H₂S Gas Detectors:

Employees and Contractors shall carry personal H₂S monitors or portable multi-gas monitors (with H₂S sensors) when working on the production jacket.

Visual & Audible Alarms:

Automatic H₂S & Combustible Gas explosion proof alarms (audible & visual) shall be located at the following locations as a minimum:

- a) **MODUs** - Drill Floor, Quarters Facing the Pipe Pack, Mud Pits, Mud Pump Room, Sack Room, SCR/Engine Room, Shale Shaker, Under Cantilever, each level of the accommodation hallways, Other normally manned worksites such as logging units/directional drilling units.
- b) **Well Service MOUs** - Each level of the accommodation hallways. Normally manned worksites out with of the accommodation, strategically placed on main deck level.

Breathing Air Cascade System:

- a) Shall be equipped with a comprehensive breathing air system that as a minimum meets the requirements of the OSHA standard ref. 29 CFR 1910.134 and the CGA bulletins No. G 7.0 and 7.1. Such breathing air systems shall provide grade D breathing air at the user outlets, as defined in the standard. These standards also specify the maintenance and care of breathing air compressors. These minimum standards shall be met at all times.
- b) The system shall comprise of Air compressors, regulators; skid mounted cascade storage bottles and strategically placed breathing air manifolds, complete with hose and stainless steel quick connect fittings that are compatible with both Company and Contractor supplied SCBAs.
- c) The design of the Breathing Air System shall be as per CGA standards or an international equivalent, by an independent registered professional engineer or engineering company approved by QatarEnergy-IS.
- d) The breathing air system cascade bottles shall have the capability to supply a minimum of one hundred (100) litres/min (3.5 cu ft/min) to each connection on the manifold breathing air system. The cascades shall be designed and sized appropriately to provide breathing air to the manifold stations for maximum number of personnel on board (POB) for not less than sixty (60) minutes.
- e) Cascade Volume Requirements - e.g. - 90 POB x 60 mins x 3.5 cu ft=18,900 cu ft Minimum Total Cascade Volume.



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High Pressure Side:

- a) Due to extreme temperature variations, consideration shall be given to thermal expansion of the gases in the system and system cascade bottles. Suitable protection shall be provided to ensure safe operation of the breathing air system. The breathing air system shall be capable of providing Grade D breathing air to the system's storage bottle cascades, from breathing air compressors operating in environments of ambient high temperatures and humidity.
- b) The system shall be capable of re-charging SCBAs to three thousand (3,000) PSI. The cascade bottle racks shall be equipped with flexible hose connections and quick fill connections capable of charging either Company or Vendor SCBA bottles after each use.
- c) Each cascade rack shall be connected to a high-pressure header, with Manual controls, allowing the transfer of compressed air between cascades. Due to this additional safety measure, no check valves shall be installed in the system other than the check valves installed as part of the various pressure regulators.
- d) Each cascade rack shall be fitted with a low-pressure alarm.
- e) The cascade racks shall be capable of being isolated manually in case of system supply line failure and capable of being individually operated to provide emergency breathing air to the respective areas.
- f) Distribution of the high pressure compressed Grade D breathing air to the high-pressure breathing air storage cascade system shall be with high-pressure stainless-steel tubing or high pressure rated anti-static hose and fittings of equivalent rating. The high-pressure tubing/hose shall be secured in place by means of stainless-steel tie wraps or equivalent at regular intervals.
- g) As a minimum, the high-pressure piping or hose shall be rated to the maximum pressure of the compressor plus ten per cent (10%).
- h) The high-pressure breathing air system and compressor shall be adequately protected from over pressure by pressure safety valves (PSVs) designed to operate at the system operating pressure plus ten per cent (10%).
- i) The Offshore units emergency control room shall be fitted with a pressure gauge that is tied into the high-pressure side of the breathing air system. This gauge will indicate overall system pressure available and will be used to determine emergency response action levels.
- j) Compressors shall be designed to provide Grade D breathing quality air at a specified rated pressure suitable to satisfy each cascade bottle's charging requirements. The main compressor shall be driven by an electric motor. A backup compressor, as capable as the main, shall be connected to the Unit's backup generator or shall be a free-standing compression engine unit. The system shall be capable of recharging all the SCBAs on site from zero psi to working pressure within two (2) hours.

Low Pressure Side:

- a) The air discharge from the cascades shall be routed through pressure-reducing/regulating valve(s) reducing the pressure to approximately 100 psi (7 Bar) or the working pressure of manifold system and/or SCBA provided. The regulating valve(s) shall be located on or adjacent to the cascade bottle rack.
- b) Each manifold header shall be constructed of stainless steel with a minimum internal diameter and of a size to service the number of outlets per header (see



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- CGA Bulletins and American Society of Mechanical Engineers (ASME) pressure vessel service).
- c) Breathing air manifold outlets shall be provided in areas where operations personnel may be required to use SCBA units, in order to carry out work tasks.
- d) Breathing air manifolds with sufficient number of breathing air outlets shall be supplied to accommodate:- max POB x 125%.
- e) Located as follows:
- f) 100% x max POB at Primary H₂S muster stations/evacuation points.
- g) 25% max POB distributed at Emergency/Control Work Areas.
- h) A pressure gauge shall be installed at each manifold location to indicate the system pressure.
- i) All cascade manifold fittings shall be stainless steel quick connect fittings, that are compatible with both Company and Contractor equipment, the type and manufacturer shall be determined during the Risk Assessment process.
- j) The low-pressure system shall be monitored by a visual and audible alarm system that alarms when the low-pressure system pressure is below 25% of its normal operating pressure.
- k) The low-pressure hoses, fittings and manifolds shall be protected from over pressurization by strategically placed PSVs set at no more than one hundred ten percent (110%) of normal system operating pressure, e.g. 110 psig on a 100 psig system.
- l) Low pressure breathing air shall be distributed from the cascade stations to the manifolds utilizing low pressure SCBA breathing air tubing or anti-static hose and fittings of the equivalent rating from the manufacturer of the SCBA in use.

Where permanent breathing air cascade systems are installed, a site-specific schematic of the breathing air system layout is to be included. (APPENDIX C)

Commissioning of Breathing Air Cascade:

- a) The breathing air system shall be pneumatically tested to one hundred and ten percent (110%) of the rated operating pressure prior to commissioning the system for use.
- b) All breathing air system pressure relief valves shall be certified before installation in the breathing air system.
- c) Each muster point shall be function tested to confirm breathing air delivery with the system fully loaded. This test shall be witnessed by a qualified QatarEnergy-IS representative.
- d) Each TEMPSC shall be function tested to confirm breathing air delivery with the system fully loaded. This test shall be witnessed by a qualified QatarEnergy-IS representative.
- e) All crews shall demonstrate competency at a fire/abandon ship drill and an H₂S evacuation drill to the satisfaction of an QatarEnergy-IS representative. It is the Contractor's responsibility to ensure that these drills are witnessed by an QatarEnergy-IS representative.

H₂S Technician

- a) Shall be trained and experienced in the maintenance, minor repair and instructing of others in the use of all breathing air system equipment provided.

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- H₂S technicians shall be certified by the equipment manufacturer to the highest maintenance level certification achievable for field maintenance
- Shall also be trained in the daily, weekly and annual care and maintenance of the breathing air compressors and shall be able to carry out testing of the stored air to ensure that it meets CGA grade D breathing air specifications.
 - Shall operate the breathing air system compressors and shall ensure that the system cascades are fully charged at all times, all cascade bottles are in the fully open position, pipe work is free of leaks and compressors are set to operate in the manual position.
 - Shall conduct Breathing air quality testing, monthly at a minimum, which satisfy the parameters set by the Compressed Gas Association in CGA G7.1 'Commodity Specification for Breathing Air' APPENDIX B.
 - Breathing air quality tests shall be conducted by sampling air at both the compressor and at the manifolds.
 - Breathing air shall only be made when it has been confirmed by tri-meter that the air supply at the compressor inlet has been tested.
 - Shall conduct weekly Bump test of the Fixed gas detection system.
 - Shall conduct Quarterly Calibration of the Fixed gas detection system.
 - (MODUs) Shall log and conduct daily/pre use function testing of personal H₂S detectors.
 - All maintenance work carried out on the breathing air cascade system and Fixed Gas Detection system shall be controlled under the installation's Permit to Work System.

TEMPSC- Breathing Air Manifolds

- The system shall supply Grade D breathing air into all the offshore units TEMPSCs by means of an umbilical type connection. The supply shall be maintained until the TEMPSC has been launched from its Davits.
- All of the offshore units TEMPSCs shall have the capability of supplying air, via individual hose supply to, one hundred percent (100%) of its seating capacity.
- The penetration for the TEMPSC cascade system umbilical shall be approved by the Survival Craft's manufacturer.

5.0 REFERENCES TO OTHER DOCUMENTS

Document Title	Document Number	Internal/ External Document	Document Referencing
Emergency Management Procedure	IS-HSE-PRC-001	Internal	Sideward
HSE Training Procedure	IS-HSE-PRC-002	Internal	Sideward
Respiratory Protection Procedure	IS-HSE-PRC-013	Internal	Sideward
American Petroleum Institute - Recommended Practices for Safe Drilling of Wells Containing Hydrogen Sulfide.	API-RP 49	External	Sideward

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Document Title	Document Number	Internal/ External Document	Document Referencing
Recommended Practice for Oil and Gas Producing and Gas Processing Plant Operations Involving Hydrogen Sulfide	API RP 55	External	Sideward
Training Requirements for Hydrogen Sulfide Safety Training	ANSI 2390.1	External	Sideward
Confined Space Entry	IS-FOP-PRC-206	Internal	Sideward
PS-1 Emergency Response Procedure	IS-FOP-PRC-210	Internal	Sideward
Halul PWHF Emergency Response Procedure	IS-FOP-PRC-410	Internal	Sideward
Offshore Pre-Incident Procedures	IS-FOP-PRC-450	Internal	Sideward
Visiting/Working on Unmanned Remote Installations	IS-FOP-PRC-204	Internal	Sideward
Marine Operations Manual	IS-SCM-PRC-001	Internal	Sideward
Procedure for Engineering & procurement	IS-FAC-PRC-224	Internal	Sideward

6.0 DEFINITIONS

Term	Definition
Hydrogen Sulfide (H ₂ S)	A flammable and poisonous gas that is heavier than air and is sometimes found in fluids encountered in oil and gas production or gas processing operations.
Immediately Dangerous to Life or Health (IDLH)	An atmospheric concentration of any toxic, corrosive or asphyxiant substance that poses an immediate threat to life or would cause irreversible or delayed adverse health effects or would interfere with an individual's ability to self-rescue/escape from a dangerous atmosphere. 100 ppm is the IDLH concentration established by NIOSH for H ₂ S.
Offshore Unit	PQL's PS1 Complex, MODUs/MOUs., Halul Island
POB	Personnel on Board (POB) shall be defined as maximum personnel on the Unit assigned a muster point or as defined by the Unit's Certifying Authority.
QUICK FILL	A system whereby the SCBA can be filled through a special fitting, from zero pressure to working pressure within 15 seconds without the removal of the bottle from the SCBA.
TLV	Threshold Limit Values -Represents an exposure level under which most people can work, day after day without adverse effects. 10 ppm

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Term	Definition
	is the TLV for H ₂ S established by NIOSH for a 40-hour work week with 8-hour days.
WORK UNIT	Self-Contained Breathing Apparatus with a minimum supply of 15 minutes of positive pressured air when disconnected for escape purposes from a cascade supply system.

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APPENDIX

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APPENDIX A - FACIAL HAIR GUIDELINES

UNACCEPTABLE



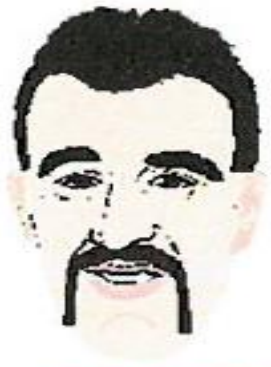
FULL BEARD



GOATEE & WIDE MUSTACHE



GOATEE & NARROW MUSTACHE



ACCEPTABLE



CLEAN SHAVEN



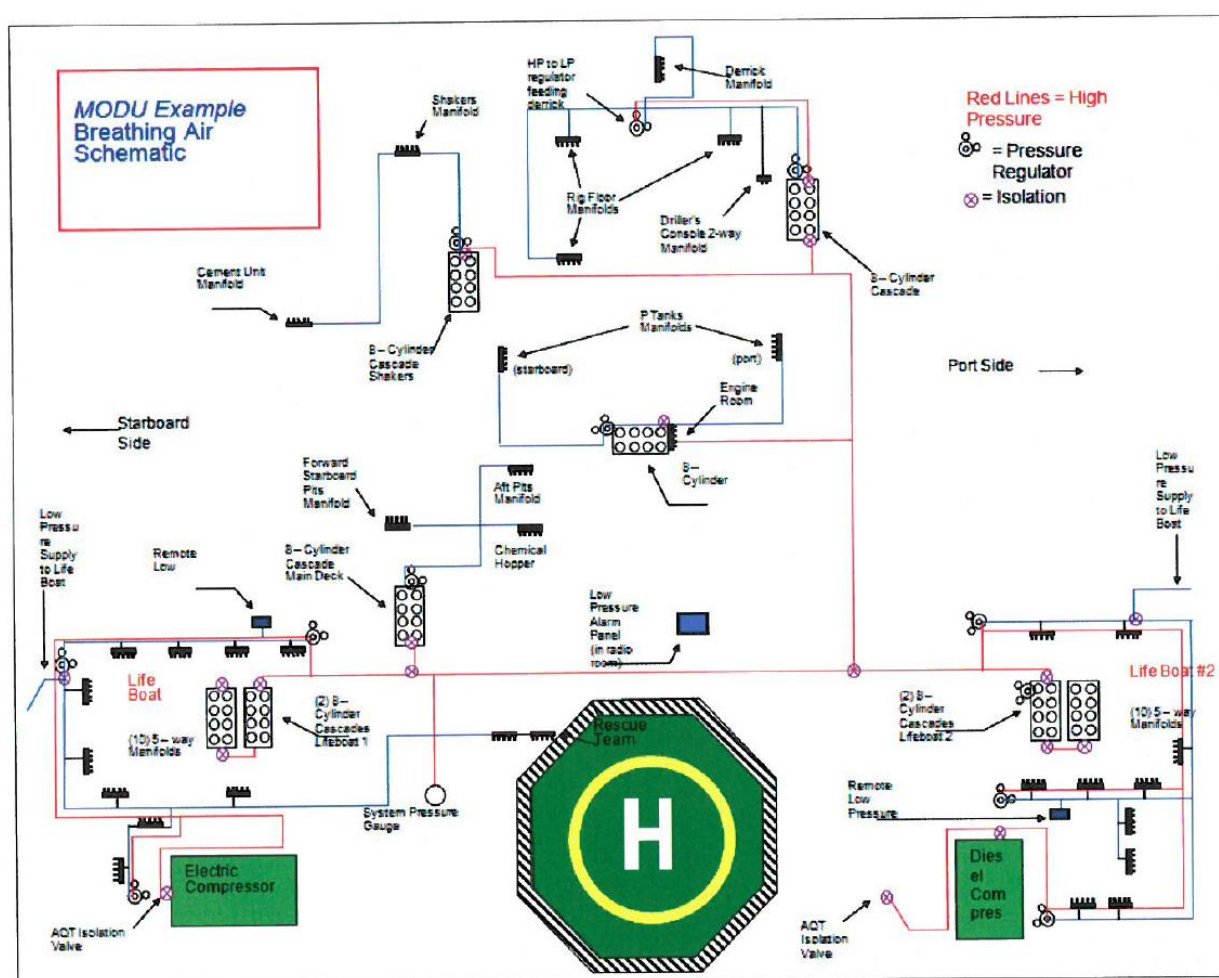
NARROW MUSTACHE

APPENDIX B - AIR QUALITY - CGA GRADE D STANDARDS

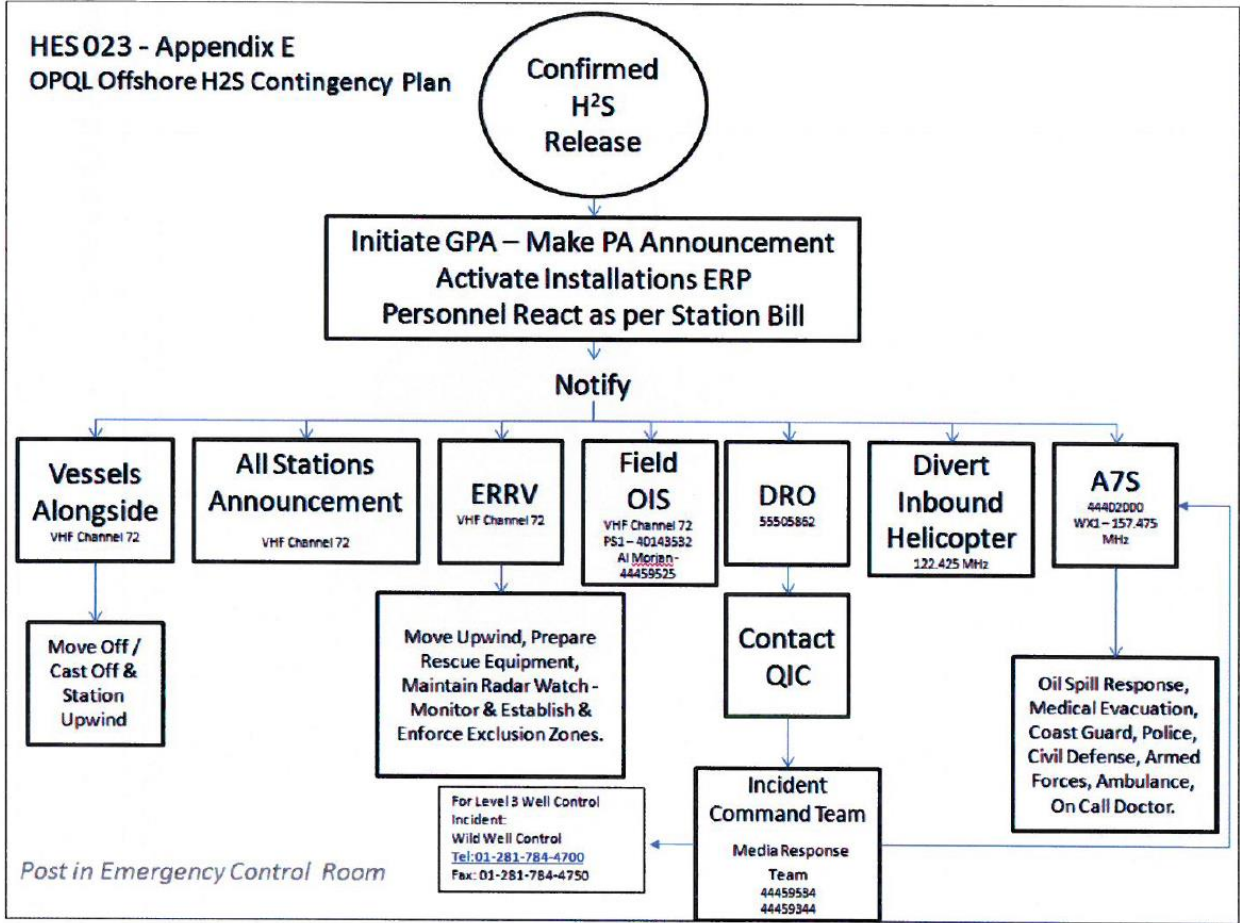
Carbon Monoxide	10 ppm
Carbon Dioxide	1000 ppm
Condensed Hydrocarbons	< 5 mg/m3
Oxygen	19.5 - 23.5
Water Vapour	*
Free Odour	

* The water content may vary depending on the use. For use with SCBA in extreme cold temperatures Dew Point should not exceed - 65 F (24ppm) or 10 degrees Fahrenheit lower than the coldest temperature expected.

APPENDIX C - MODU EXAMPLE BREATHING AIR SCHEMATIC



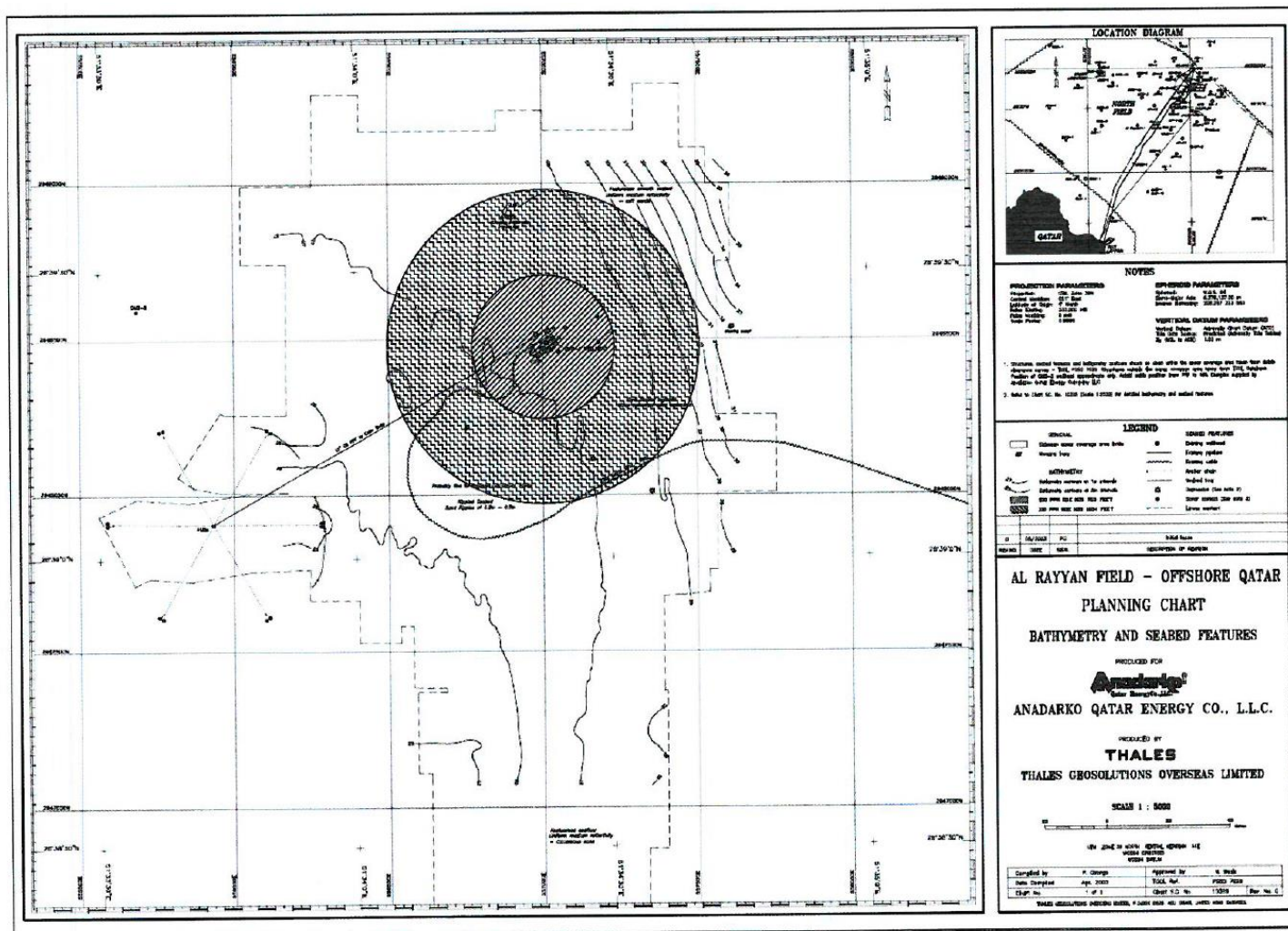
APPENDIX D – OFFSHORE H2S CONTINGENCY PLAN



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APPENDIX E - RADIUS OF EXPOSURE



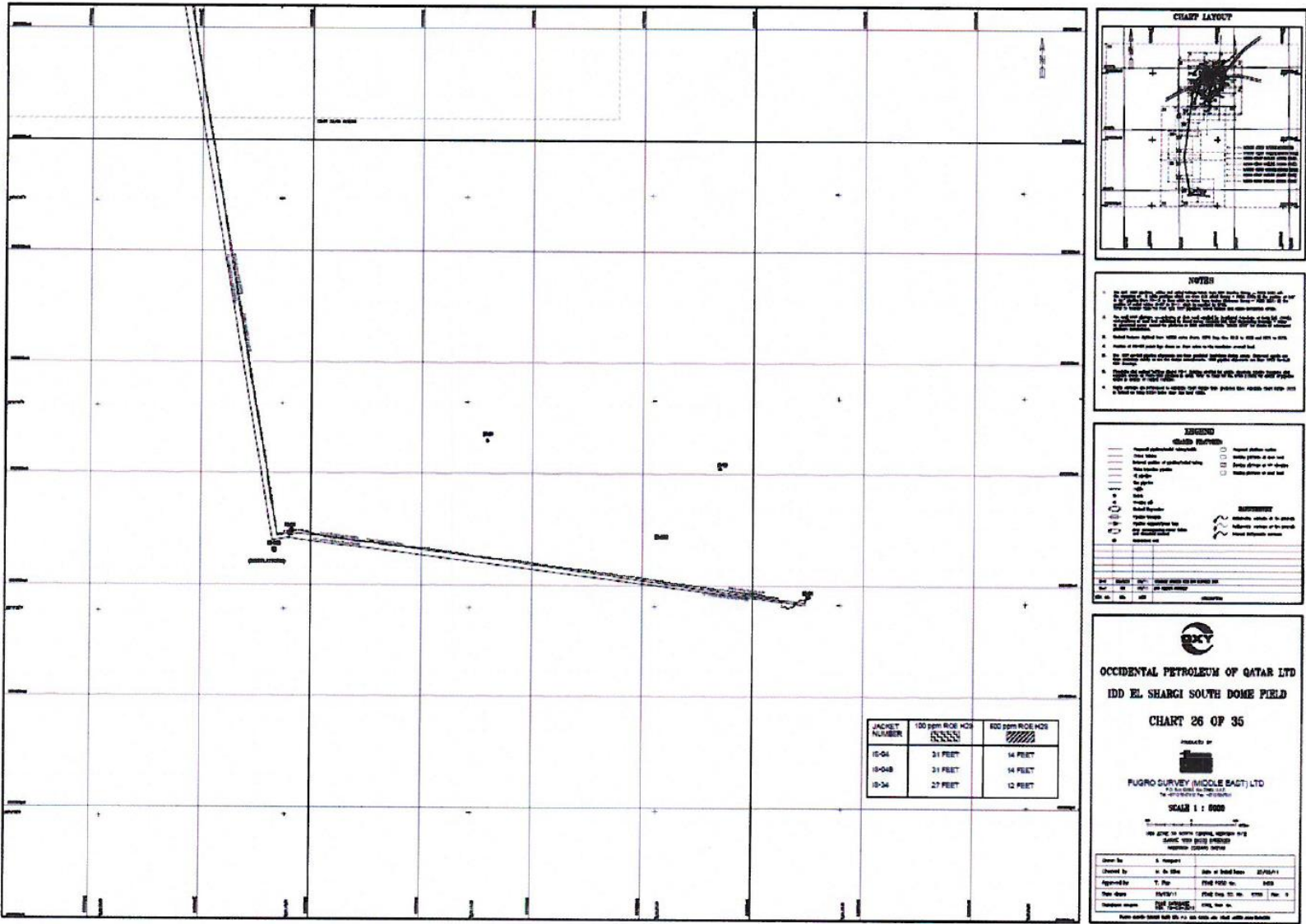


CLASSIFICATION: C1 - CONTROLLED

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DOC NO: IS-HSE-PRC-023

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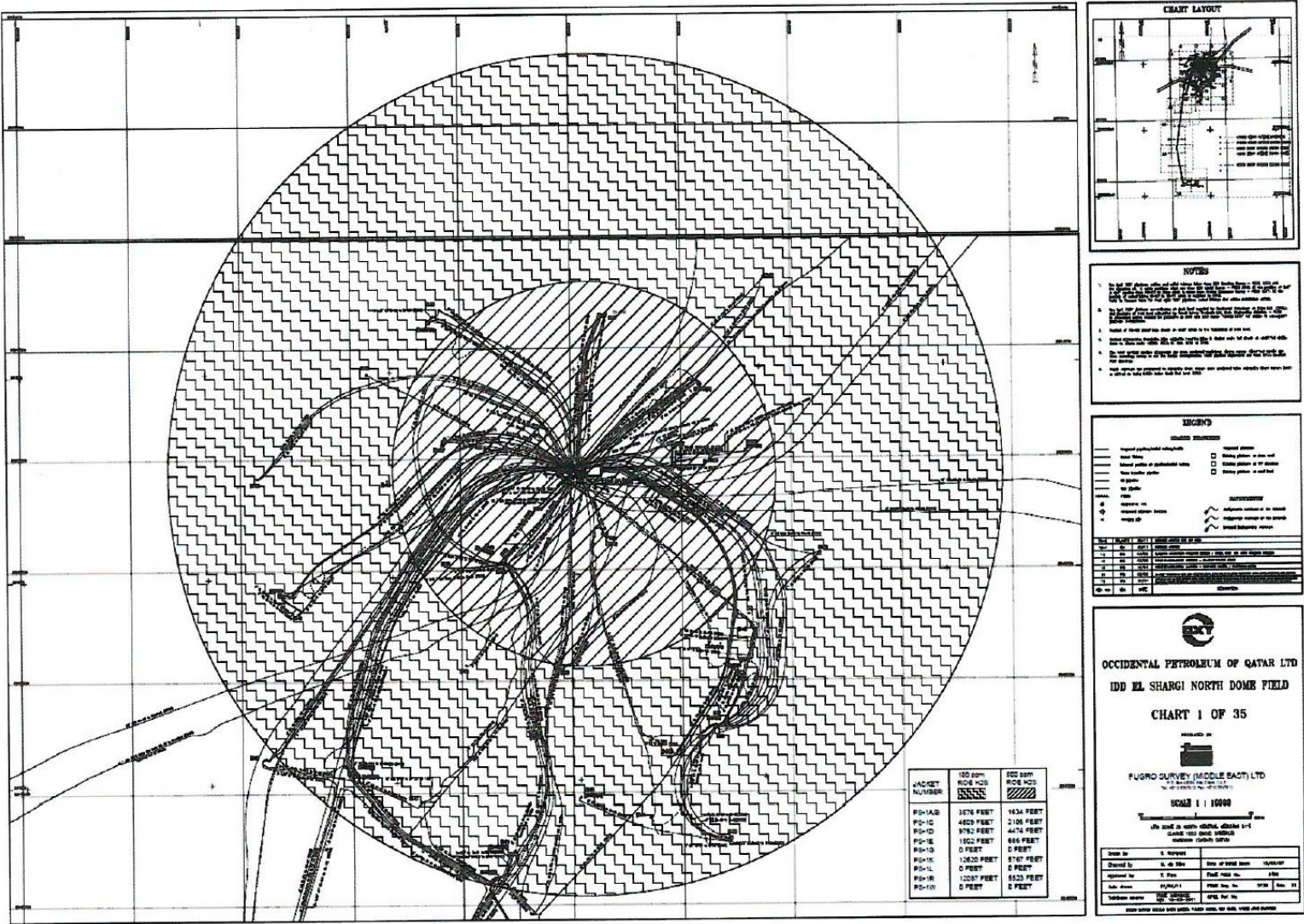


CLASSIFICATION: C1 - CONTROLLED

IS HYDROGEN SULFIDE SAFETY PROCEDURE

DOC NO: IS-HSE-PRC-023

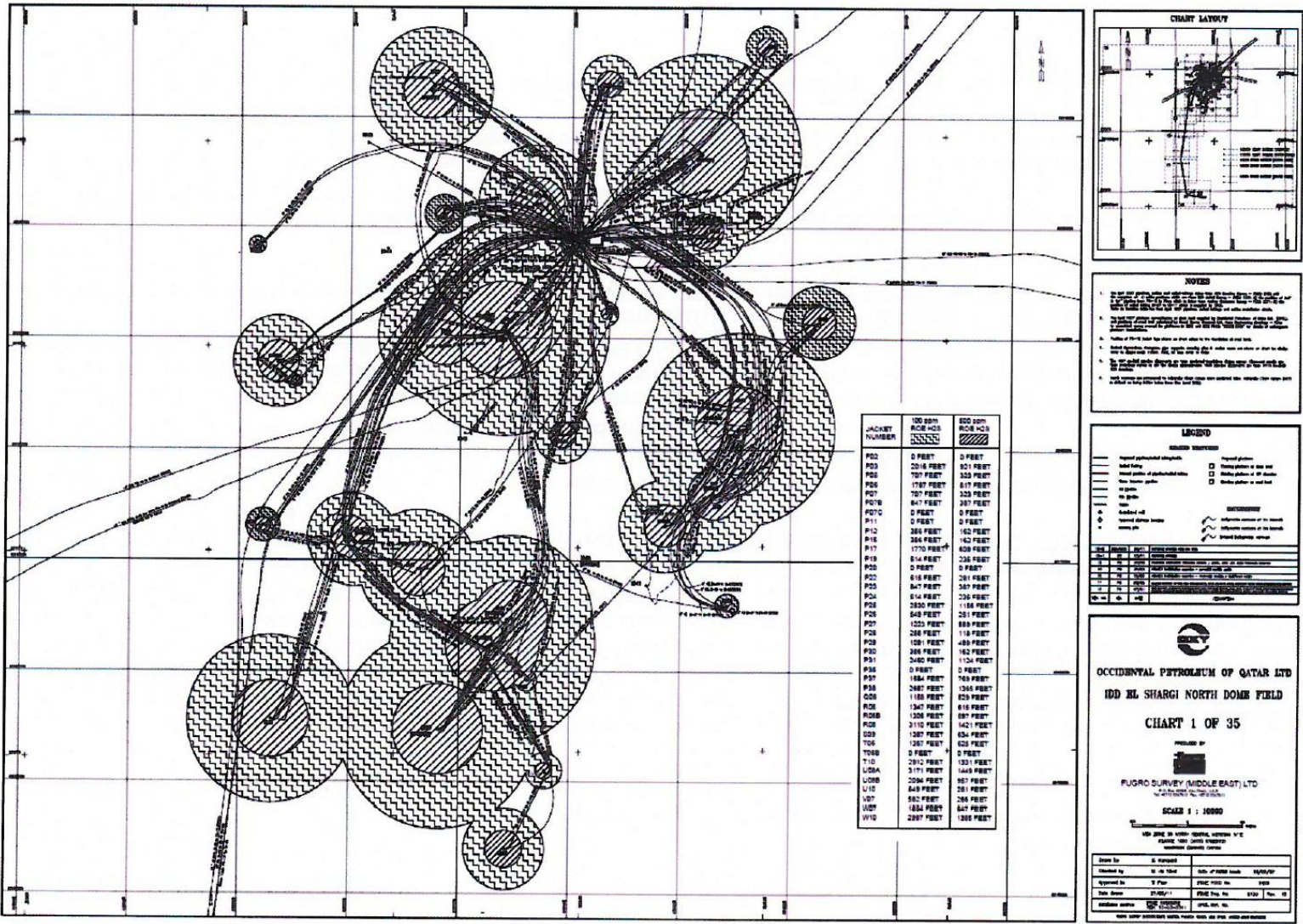
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APPENDIX F - BASIC PROPERTIES OF H₂S

Basic Properties of H₂S

H₂S is very toxic and can cause fatalities. A single breath, at a concentration of 1000 ppm, may cause the victim to become unconscious and to stop breathing within a couple of minutes.

H₂S is colorless.

H₂S has the odor of rotten eggs at low concentrations. The odor threshold for H₂S is 0.3 ppm. At concentrations above 100 ppm, H₂S paralyses the olfactory nerve eliminating the sense of smell. For this reason, the sense of smell cannot be trusted to detect H₂S. Therefore, all QatarEnergy-IS facilities shall use personal, portable and fixed gas detectors as appropriate.

H₂S is heavier than air, with a vapor density (specific gravity) of 1.189 relative to air. Thus, H₂S will tend to collect in low lying areas and may concentrate to dangerous levels. If H₂S is mixed with other gases and released under elevated temperatures or pressures, H₂S may initially be carried upward and either be dispersed or may settle downrange of the release.

The auto-ignition temperature for H₂S is very low, 500°F/260°C. If burned, it produces a blue flame and produces Sulfur Dioxide (SO₂). SO₂ is also toxic, irritating to the eyes and lungs, and can cause chemical pneumonia after a few hours of exposure. SO₂ is an irritant at very low levels and requires the use of RPE at a level of 2 ppm.

H₂S is flammable from 4.3% - 46% by volume in air. This range of flammability is greater than hydrocarbon vapors in natural gas.

H₂S is soluble in water and most liquid hydrocarbons. When mixed with water, it produces sulfurous acid. This acid at concentrations greater than 100 ppm causes irritation to the eyes, throat, respiratory tract and lungs.

When H₂S is in aqueous solution, H₂S causes general corrosion through chemical reactions with the materials in the metal. H₂S under specific conditions and concentrations can cause hydrogen embrittlement or sulfide stress cracking. For elevated concentrations of H₂S, proper metallurgies shall be determined for use as per NACE Standard MR-01-75 (latest edition).

H₂S in the presence of ferretic steel will form a corrosion product of iron sulfide. This material is pyrophoric and may react violently when exposed to oxygen. To prevent fire potential, iron sulfide scale shall be kept wet or not be brought in to contact with oxygen.

H₂S Effects on the Body

The principal hazard of H₂S inhalation is death caused by paralysis of the respiratory system. The inhaled gas is absorbed into the bloodstream and is then carried to the brain where it affects the respiratory nerve centre.

Other symptoms of H₂S exposure include headaches, dizziness, drowsiness, increased heart rate, and nausea, with severity being determined by the amount of exposure. Coughing and pain in the eyes, throat, and chest may be attributed to the formation of acid formed when H₂S comes into contact with the moist surfaces of body tissue.

The rapidity and impact of H₂S exposure to personnel are predicated on four main variables. These are:

- Duration

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- Concentration
- Physical conditioning of the individual.
- Frequency and history of individual's exposures.

Short term repeated exposure to elevated concentrations of H₂S has a cumulative effect, and care shall be taken to ensure that time without exposure is given to the individual to allow residual gas to be flushed from the body (some studies have indicated this could take as long as 120 days to completely remove traces of H₂S from the blood stream). Anecdotal evidence has shown that personnel exposed to high concentrations of H₂S have permanent lower resistance to H₂S. Alcohol in the bloodstream greatly reduces resistance to H₂S by causing H₂S to be absorbed much more quickly than normal.

Exposure Limits

QatarEnergy IS recognizes that the PEL for H₂S is 10 ppm for an 8-hour work day.

APPENDIX G – REVISION HISTORY LOG**Revision Number: 02****Date: 24/03/2024**

Item Revised	Revision Description	Page No.
Appendix	I. Appendix A - Equipment Matrix removed.	Appendix
4.2	II. Section on Reaction Plan removed.	4
4.4	III. FIGURE 1 amended to state 'Don EBA' upon alarm activation.	6
4.5	IV. H ₂ S Training changed from OPITO Approved to OPITO/QatarEnergy Approved.	6
4.5	V. H ₂ S Training Validity changed to 2-years for QE staff.	6
4.12.4	VI. EEBA quantity changed from 150% to 125% of the personnel on board.	15
4.12.1	VII. Personal H ₂ S monitors to be worn by all personnel, not just core crew.	12

Remarks :**APPENDIX H – LIST OF ABBREVIATIONS**

Term	Definition
ANSI	American National Standard Institute
API	American Petroleum Institute
BA	Breathing Apparatus or SCBA
CFR	Congressional Federal Register
CGA	Compressed Gas Association (USA)
CO ₂	Carbon Dioxide

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Term	Definition
CPR	Cardio Pulmonary Resuscitation
DSM	Drill Site Manager
DST	Drill Stem Testing/Tools
EN	European Norms
GPA	General Platform Alarm
MODU	Mobile Offshore Drilling Unit
MOU	Mobile Offshore Unit
NACE	National Association of Corrosion Engineers
NIOSH	National Institute of Occupational Safety and Health (USA)
O ₂	Oxygen
QatarEnergy-IS	Idd El-Shargi Asset of QatarEnergy
OSHA	Occupational Safety and Health Administration
PEL	Permissible Exposure Limit
PPM	Part Per Million by volume in air
PSI	Pound per Square Inch
RPE	Respiratory Protective Equipment
SABA	Supplied Air Breathing Apparatus
SBA	Safe Briefing Area
SCBA	Self-Contained Breathing Apparatus
Shall	Mandatory
SO ₂	Sulfur Dioxide
TEMPSC	Totally Enclosed Motorized Propelled Survival Craft